The Beauty of Mathematics: From Algebra to Geometry

FAN, Sixing

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Preface: Mathematical Discovery Arises from Questions

When I first started university math courses, I always had a question: how can things like the irrational numbers $\sqrt{2}$, π, and the imaginary number i exist? Especially i, which doesn't even exist in our daily lives. However, Augustin-Louis Cauchy and other mathematicians actually analyzed complex functions on a plane that doesn't exist in the real world. Are they out of their minds? But later I discovered that results in the complex domain could beautifully serve the real domain, for instance, using theorems of complex functions to compute real integrals that are otherwise hard to handle. The origins of irrational and complex numbers likely stem from two similar lines of inquiry.

We know that an integer N multiplied by itself is still an integer and a positive one, called a perfect square. What if a number x multiplied by itself is not a perfect square? Further, what if a number y multiplied by itself is not positive but -1? The questions behind these are precisely the two important number systems in mathematics: irrational and imaginary numbers.

So you see, mathematical discovery often stems from inquiry. The journey from inquiry to major discovery isn't easy, but I want to say that your whimsical ideas, logical reasoning, and mathematical reflections today might contain the seeds of significant discoveries. The following content presents some of my reflections on university mathematics. I aim to share the marvelous aspects I've discovered in mathematics using natural and accessible language, hoping to inspire the reader.

Chapter 1

Construction of the Real Number System—Analysis

1.1 Thoughts Inspired by Natural Numbers

Human understanding of numbers began with natural numbers, the positive integers we used to count sheep as children:

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

That is, when quantifying discrete objects, people naturally use the concept of positive integers. However, there seems to be a slight issue. Everything in our world exists in finite quantities: the population of Earth, the amount of currency, the number of celestial bodies. These numbers may be large, but there is always a number that represents them. In other words, if you count from 1, you will eventually stop at that number. But, **is there a largest positive integer**?

This question seems easy to answer. Suppose, on the contrary, that there exists a largest positive integer, called N. Then N+1 is also a positive integer and is larger than N. Therefore, we say there is no largest positive integer. We call this imagined infinitely large quantity positive infinity.

Positive infinity cannot actually be compared in size, but we can try to categorize it into different levels based on properties. This process leads to many interesting questions, which I will elaborate on in Section 3.

1.2 From Natural Numbers to Rational Numbers

When we need to solve the equation $2x = 1$, natural numbers are insufficient, giving rise to the fraction/rational number system:

$$\mathbb{Q} = \left\{ \frac{p}{q} \mid p \in \mathbb{Z}, q \in \mathbb{Z}^* \right\}$$

The rational number system seemed complete, but when facing $x^2 = 2$, the Pythagoreans discovered that $\sqrt{2}$ could not be expressed as a ratio of two integers. This was the first crisis in the history of mathematics and a major turning point—the discovery of irrational numbers.

1.3 Axiomatic Construction of Real Numbers

The foundation of the mathematical edifice is a set of axioms; everything else is logic. For instance, the real number system can be characterized by a set of axioms:

- **Field Axioms**:
  The set of real numbers $\mathbb{R}$ forms a field under addition and multiplication.
- **Order Axioms**:
  $\mathbb{R}$ is a totally ordered set, and the order is compatible with the operations.
- **Completeness Axiom** (key):
  Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound.

Completeness is the essential difference between real and rational numbers. It has multiple equivalent formulations:

$$\begin{aligned}
\text{Supremum Principle} &\Leftrightarrow \text{Monotone Convergence Theorem} \\
&\Leftrightarrow \text{Nested Intervals Theorem} \\
&\Leftrightarrow \text{Heine–Borel Theorem (Finite Covering Property)} \\
&\Leftrightarrow \text{Cauchy Convergence Criterion}
\end{aligned}$$

Chapter 2

Modern Systems for Expressing Numbers—Base

2.1 Why Decimal?

We know that in arithmetic, we cannot create an infinite number of symbols. Therefore, numbers need a positional notation system, where a ”1” in the next position represents the ”maximum” of the previous position. Why do humans commonly use the decimal system? A natural explanation is: **humans have ten fingers**. After counting ten fingers, we ”carry over” and use a new position to represent larger quantities.

2.2 Applications of Different Bases

- **Binary** ($b = 2$): The foundation of computer science; machine code is a sequence of “0” and “1”.
- **Base-60** ($b = 60$): A traditional system for measuring angles and time, e.g., 1 hour = 60 minutes, 1 minute = 60 seconds.
- **Hexadecimal** ($b = 16$): Compact representation in computer programming.

2.3 Mathematical Principle of Positional Notation

To distinguish between the actual magnitude of a number and its base-b representation, we express the representation for any base b ($b \geq 2$):

$$a_n a_{n-1} \cdots a_1 a_{0(b)} = a_n b^n + a_{n-1} b^{n-1} + \cdots + a_1 b + a_0$$

where $0 \leq a_i < b$.

Chapter 3

Beyond Finite Human Imagination—Limits

3.1 Where Limits Are Useful

In practice, we always hope to solve problems in a finite number of steps. Therefore, mathematics does not welcome "infinite" processes. However, limits can help us address existence problems. For example: **Bisection Method for Finding Roots** Consider a continuous function $f(x)$ on $[a, b]$ satisfying $f(a)f(b) < 0$.

1. Take the midpoint $c = \frac{a+b}{2}$
2. If $f(c) = 0$, the root is found.
3. If $f(a)f(c) < 0$, continue in $[a, c]$
4. Otherwise, continue in $[c, b]$

If the root is found in a finite number of steps, the problem is solved. If not found finitely, we can prove that the limit of the sequence $\{c_n\}$ is the root:

$$\lim_{n\to\infty} c_n = \xi, \quad f(\xi) = 0$$

Note: **Rigorous Definition of Limit** The Cauchy–Weierstrass ϵ-δ definition:

$$\lim_{x\to a} f(x) = L \iff \forall\epsilon > 0, \exists\delta > 0 : 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \epsilon$$

This is a classic example of proving existence via limits.

3.2 Hilbert's Hotel: The Marvelous World of Infinity

This thought experiment, proposed by German mathematician David Hilbert in 1924, vividly illustrates the counterintuitive nature of the concept of "infinity" and reveals the mathematical principles behind limit thinking.
Imagine a hotel with **infinitely many rooms** (numbered: 1, 2, 3, ...). One day, all rooms are occupied. So how do you accommodate new guests?

Case 1: **A single new guest arrives**:
The manager asks the guest in **Room 1** to move to **Room 2**, the guest in **Room 2** to move to **Room 3**, and so on. Each current guest moves from room n to room $n + 1$. This vacates Room 1, and the new guest can check in.

Case 2: **A bus with infinitely many guests arrives**:
The manager asks the current guest in room n to move to room $2n$ (i.e., 1→2, 2→4, 3→6, ...). This vacates all **odd-numbered rooms** (1, 3, 5, ...), and the infinitely many new guests can check in sequentially.

Case 3: **Infinitely many buses, each carrying infinitely many guests**:
The manager can cleverly utilize the infinity of prime numbers. He asks the current guest in room n to move to room 2^n. Then, he assigns the n-th passenger on the m-th bus to the room numbered p_m^n (where p_m is the m-th prime). This method uniquely assigns a room number to each new guest.

In a finite world, "all rooms are full" means that no one else can be accommodated. The beauty of this problem lies in the realm of "infinity": **"fully occupied" and "having space" can coexist**. This idea is frequently used in mathematical analysis.

3.3 Countable Infinity and One-to-One Correspondence

Hilbert's Hotel is not a true logical paradox but reveals the essential nature of "countably infinite" sets:

1. **Equipotence**: The set of natural numbers $\mathbb{N} = \{1, 2, 3, \ldots\}$ can be put into a one-to-one correspondence with a proper subset of itself (e.g., the set of even numbers):

   $$f : \mathbb{N} \to \{\text{even numbers}\}, \quad f(n) = 2n$$

   Mathematically, this means: **A part can equal the whole**—the most fundamental property of infinite sets.

2. **Mathematical Operation**: The hotel manager's operations essentially construct new "one-to-one correspondences":

   - Accommodating finitely many new guests: $\mathbb{N} \cup \{\text{finite set}\} \sim \mathbb{N}$
   - Accommodating one bus of guests: $\mathbb{N} \cup \mathbb{N} \sim \mathbb{N}$
   - Accommodating infinitely many buses: $\mathbb{N} \times \mathbb{N} \sim \mathbb{N}$

The story of Hilbert's Hotel tells us that the idea of limits is not just a mathematical tool but an important way of thinking that allows humans to break through finite cognition and explore the infinite world. Just as the hotel can always accommodate new guests, mathematical thinking can always open new rooms for understanding the world.

Chapter 4

A Detour for Real Integrals—Complex Numbers

4.1 Difficulties with Real Integrals

We know that computing real integrals typically requires finding antiderivatives. Common methods include:

1. Substitution
2. Integration by parts
3. Partial fraction decomposition

But some integrals are difficult to compute within real analysis, such as:

$$\int_0^\infty \frac{\sin x}{x}\,dx, \quad \int_{-\infty}^\infty e^{-x^2}\,dx$$

4.2 Application of Complex Numbers in Integration

Recall the general form of a complex number: $z = a + bi$, where $a, b \in \mathbb{R}$. We know many real integrals are hard to handle directly, but complex integration can detour to solve complicated real integrals. As the famous mathematician said:

> The shortest path between two truths in the real domain passes through the complex domain.
>
> — Jacques Hadamard

The following two results are our secret weapons for solving real integrals via detours.

Euler's Formula

$$e^{ix} = \cos x + i \sin x$$

This formula bridges trigonometric and exponential functions.

Residue Theorem

For integrals along closed contours:

$$\oint_C f(z)\,dz = 2\pi i \sum \mathrm{Res}(f, z_k)$$

where z_k are the singularities of f inside C.

Example When computing $\int_0^\infty \frac{\sin x}{x}\,dx$, consider the complex function $f(z) = \frac{e^{iz}}{z}$. Using the Residue Theorem yields the result $\frac{\pi}{2}$.

4.3 Complex Plane: Visual Demonstration of Contour Integration

Consider the real integral

$$I = \int_{-\infty}^{\infty} \frac{x^4}{x^6 - 1}\, dx.$$

Consider the corresponding principal value integral

$$I = \text{P.V.} \int_{-\infty}^{\infty} \frac{x^4}{x^6 - 1}\, dx.$$

The integrand has singularities on the real axis at the real roots of the denominator $x^6 - 1 = 0$, i.e., $x = \pm 1$.

Complex Extension and Singularity Analysis

Extend the integrand to the complex plane:

$$f(z) = \frac{z^4}{z^6 - 1}.$$

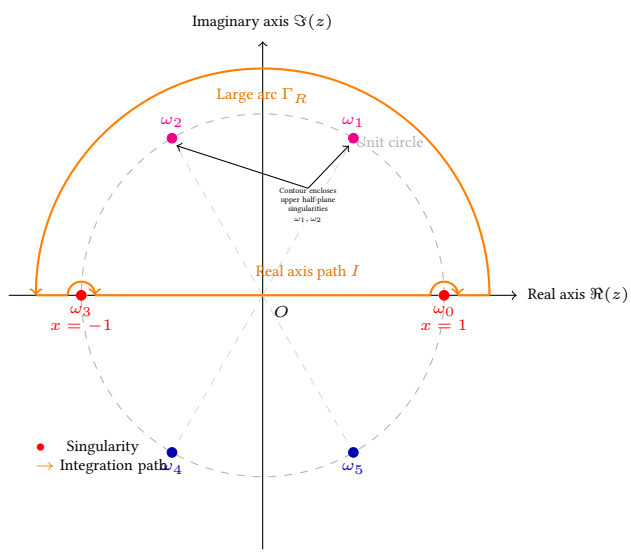
The zeros of the denominator are the six roots of unity $z^6 = 1$:

$$\omega_k = e^{i\pi k/3}, \quad k = 0, 1, \ldots, 5.$$

Specifically:

$$\omega_0 = 1,\ \omega_1 = e^{i\pi/3},\ \omega_2 = e^{i2\pi/3},\ \omega_3 = -1,\ \omega_4 = e^{i4\pi/3},\ \omega_5 = e^{i5\pi/3}.$$

Here $\omega_0 = 1$, $\omega_3 = -1$ lie on the real axis; ω_1, ω_2 are in the upper half-plane; ω_4, ω_5 are in the lower half-plane.



Contour Selection and Computation

Choose the closed contour C as shown, consisting of:

- The path along the real axis from $-R$ to R, but detouring around $x = \pm 1$ via small semicircles in the upper half-plane.
- A large semicircular arc Γ_R of radius R in the upper half-plane.

1. Residue Calculation For a simple pole ω_k:

$$\operatorname{Res}(f, \omega_k) = \frac{\omega_k^4}{6\omega_k^5} = \frac{1}{6\omega_k}.$$

The singularities enclosed by C in the upper half-plane are ω_1 and ω_2:

$$\operatorname{Res}(f, \omega_1) = \frac{1}{6e^{i\pi/3}}, \quad \operatorname{Res}(f, \omega_2) = \frac{1}{6e^{i2\pi/3}}.$$

Summing:

$$\sum_{\text{upper}} \operatorname{Res} = \frac{1}{6}(e^{-i\pi/3} + e^{-i2\pi/3}) = \frac{1}{6}\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i - \frac{1}{2} - \frac{\sqrt{3}}{2}i\right) = -\frac{\sqrt{3}}{6}i.$$

2. Small Semicircle Integrals Integrals over the small semicircles at $z = \pm 1$ (counterclockwise in upper half-plane):

$$\int_{\gamma_\rho^\pm} f(z)\, dz = i\pi \operatorname{Res}(f, \pm 1).$$

Compute:

$$\operatorname{Res}(f, 1) = \frac{1}{6}, \quad \operatorname{Res}(f, -1) = \frac{1}{6(-1)} = -\frac{1}{6}.$$

Thus:

$$\int_{\gamma_\rho^+(1)} f(z)\, dz = \frac{i\pi}{6}, \quad \int_{\gamma_\rho^+(-1)} f(z)\, dz = -\frac{i\pi}{6}.$$

The two small semicircle integrals cancel each other.

3. Large Arc Integral When $|z| = R$, $|f(z)| \sim R^4/R^6 = 1/R^2$. By Jordan's Lemma:

$$\lim_{R\to\infty} \int_{\Gamma_R} f(z)\, dz = 0.$$

4. Applying the Residue Theorem

$$\oint_C f(z)\, dz = 2\pi i \sum_{\text{upper}} \operatorname{Res} = 2\pi i \left(-\frac{\sqrt{3}}{6}i\right) = \frac{\pi\sqrt{3}}{3}.$$

Also,

$$\oint_C f(z)\,dz = \text{P.V.}\int_{-\infty}^{\infty} f(x)\,dx + \int_{\Gamma_R} f(z)\,dz + \sum \text{small semicircle integrals.}$$

Taking the limit $R \to \infty$, we obtain:

$$\boxed{\text{P.V.}\int_{-\infty}^{\infty} \frac{x^4}{x^6-1}\,dx = \frac{\pi\sqrt{3}}{3}.}$$

Conclusion

An integral difficult to compute on the real axis can, by extending to the complex plane and choosing a suitable closed contour, be transformed via the Residue Theorem into a sum of residues at a finite number of singularities. This is a concrete example of the detour tactic for solving real integrals.

Chapter 5

Two Perspectives on Matrix Multiplication—Insights

5.1 Standard Perspective: Row-Vector Times Column-Vector

This is the definition given in many textbooks: each element c_{ij} is determined by the i-th row of A and the j-th column of B.

Let $A \in \mathbb{R}^{m\times k}$, $B \in \mathbb{R}^{k\times n}$. The product $C = AB \in \mathbb{R}^{m\times n}$ has elements:

$$c_{ij} = \sum_{t=1}^{k} a_{it}b_{tj} = A[i,:] \cdot B[:,j]$$

where:

- $A[i,:]$ denotes the i-th row vector of A
- $B[:,j]$ denotes the j-th column vector of B
- $\cdot$ denotes the vector inner product

In matrix form:

$$\begin{aligned} C = AB &= \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mk} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{k1} & b_{k2} & \cdots & b_{kn} \end{bmatrix} \\ &= \begin{bmatrix} \sum_{t=1}^{k} a_{1t}b_{t1} & \sum_{t=1}^{k} a_{1t}b_{t2} & \cdots & \sum_{t=1}^{k} a_{1t}b_{tn} \\ \sum_{t=1}^{k} a_{2t}b_{t1} & \sum_{t=1}^{k} a_{2t}b_{t2} & \cdots & \sum_{t=1}^{k} a_{2t}b_{tn} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{t=1}^{k} a_{mt}b_{t1} & \sum_{t=1}^{k} a_{mt}b_{t2} & \cdots & \sum_{t=1}^{k} a_{mt}b_{tn} \end{bmatrix} \end{aligned}$$

Key Observation: Specifically, for each $t = 1, \ldots, k$:

- Uses element a_{it} from row i, column t of A
- Uses element b_{tj} from row t, column j of B

5.2 Decomposition Perspective: Column-Vector Times Row-Vector

Matrix multiplication $C = AB$ also admits a second interpretation: it can be expressed as the sum of k rank-1 matrices (outer products):

$$AB = \sum_{t=1}^{k} (A[:,t])(B[t,:])$$

where:

- $A[:,t] \in \mathbb{R}^m$ is the t-th column vector of A (an $m \times 1$ matrix)
- $B[t,:] \in \mathbb{R}^n$ is the t-th row vector of B (a $1 \times n$ matrix)
- $(A[:,t])(B[t,:])$ is the outer product, yielding an $m \times n$ matrix

Specifically, for each $t = 1, \ldots, k$:

$$M_t = A[:,t] \cdot B[t,:] = \begin{bmatrix} a_{1t} \\ a_{2t} \\ \vdots \\ a_{mt} \end{bmatrix} \begin{bmatrix} b_{t1} & b_{t2} & \cdots & b_{tn} \end{bmatrix} = \begin{bmatrix} a_{1t}b_{t1} & a_{1t}b_{t2} & \cdots & a_{1t}b_{tn} \\ a_{2t}b_{t1} & a_{2t}b_{t2} & \cdots & a_{2t}b_{tn} \\ \vdots & \vdots & \ddots & \vdots \\ a_{mt}b_{t1} & a_{mt}b_{t2} & \cdots & a_{mt}b_{tn} \end{bmatrix}$$

Thus:

$$C = AB = \sum_{t=1}^{k} M_t$$

Why No Cross-Multiplication?

From the first perspective, to compute c_{ij} we need:

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ik}b_{kj}$$

Note a_{ik} is the element at row i, column k of A, and b_{2j} is the element at row k, column j of B. When multiplying them, the column index of A (k) matches the row index of B (k). We discover the **matching rule** of matrix multiplication: the column index of A must match the row index of B.

Illustrative Examples

Consider the $(1,1)$ element of AB:

$$c_{11} = a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31}$$

In this sum, a_{12} (row 1, column 2 of A) multiplies b_{21} (row 2, column 1 of B). Note the "2" in a_{12} and b_{21} match. Column 2 of A does not multiply with column 1 of B.

Consider the $(2,3)$ element of AB:

$$c_{23} = a_{21}b_{13} + a_{22}b_{23} + a_{23}b_{33}$$

Here a_{22} (row 2, column 2 of A) multiplies b_{23} (row 2, column 3 of B), not any element from column 1 of B.

Summary

Matrix multiplication AB has two equivalent interpretations:

1. **Row-Column Perspective**: Each element c_{ij} is the inner product of the i-th row of A and the j-th column of B.

2. **Column-Row Perspective**: The entire matrix is the sum of k outer products, each formed by multiplying a column of A with the corresponding row of B.

The t-th column of A only multiplies with the t-th row of B, never with other rows of B. This ensures the **matching** of indices and is the basis for the equivalence of the two perspectives.

5.3 Proof of Equivalence of the Two Perspectives

To prove:

$$\sum_{t=1}^{k}(A[:,t])(B[t,:]) = AB$$

Check the (i,j) element:

$$\left[\sum_{t=1}^{k}(A[:,t])(B[t,:])\right]_{ij} = \sum_{t=1}^{k}\left[(A[:,t])(B[t,:])\right]_{ij}$$

And

$$\left[(A[:,t])(B[t,:])\right]_{ij} = a_{it}b_{tj}$$

So:

$$\left[\sum_{t=1}^{k}(A[:,t])(B[t,:])\right]_{ij} = \sum_{t=1}^{k}a_{it}b_{tj} = (AB)_{ij}$$

Therefore, the two perspectives are equivalent (for reference).

5.4 Geometric Interpretation

- **First Perspective**: The j-th column of AB is the representation of the j-th column of B in the column space of A:

$$AB[:,j] = A \cdot B[:,j] = b_{1j}A[:,1] + b_{2j}A[:,2] + \cdots + b_{kj}A[:,k]$$

- **Second Perspective**: AB is the sum of k rank-1 matrices, each spanned by a column of A and a row of B:

$$AB = \underbrace{A[:,1]B[1,:]}_{\text{spanned by col. 1 of A}} + \underbrace{A[:,2]B[2,:]}_{\text{spanned by col. 2 of A}} + \cdots + \underbrace{A[:,k]B[k,:]}_{\text{spanned by col. k of A}}$$

5.5 Significance of the Decomposition Perspective

The decomposition perspective is particularly useful in:

1. **Low-Rank Approximation**: If k is large but the matrix is approximately low-rank, the first few components can approximate the matrix.

2. **Matrix Factorization**: Singular Value Decomposition (where r is the rank of the matrix)

$$A = U\Sigma V^\top = \sum_{i=1}^{r} \sigma_i u_i v_i^\top$$

3. **Computational Efficiency**: This representation can reduce storage and computation costs when the matrix is low-rank.

4. **Understanding Linear Transformations**: Decomposing a linear transformation into a sum of rank-1 transformations.

5.6 Personal Commentary

Finally, I'd like to share some thoughts on linear algebra. In the mathematical world, there are many problems, both linear and nonlinear. However, linear problems have many methods that make them particularly amenable to treatment. Often, we can try to transform problems into linear spaces to solve them. That's why university mathematics spends considerable time teaching linear algebra. At the same time, matrices in linear algebra help us analyze high-dimensional problems, such as multivariable functions, multi-sample datasets, etc. You can view it as a tool that allows analysis of higher-dimensional situations in one step, starting from one-dimensional cases.

5.6.1 Appendix · On Matrix Rank Inequalities

1. **Rank Inequality for Matrix Sum**:

$$\text{Rank}(B + C) \leq \text{Rank}(B) + \text{Rank}(C)$$

2. **Matrix Product Rank Inequality 1**: Let $BC \in \mathbb{R}^{m\times n}$ be the product of matrices $B \in \mathbb{R}^{m\times k}$ and $C \in \mathbb{R}^{k\times n}$. Then we have

$$\text{Rank}(BC) \leq k$$

3. **Matrix Product Rank Inequality 2**:

$$\text{Rank}(BC) \leq \min\{\text{Rank}(B), \text{Rank}(C)\}$$

4. **Sylvester's Rank Inequality**: For $B \in \mathbb{R}^{m\times k}$ and $C \in \mathbb{R}^{k\times n}$,

$$\text{Rank}(B) + \text{Rank}(C) - k \leq \text{Rank}(BC) \leq \min\{\text{Rank}(B), \text{Rank}(C)\}$$

Rank inequalities are an elegant and powerful tool in linear algebra. These inequalities not only reveal how matrix operations affect the rank—an important invariant—but also embody a profound idea in mathematics: by establishing inequality relations, we can gain important insights into the structure of a problem without exact computation.

The rank of a matrix tells us the "effective dimension" of linearly independent information. Inequality 2 intuitively shows that when transmitting information through linear transformations, the "purity" or number of independent components does not increase and may even decrease, similar to the data processing inequality in information theory.

It is this structural understanding abstracted from specific computations that makes linear algebra an indispensable language and tool in modern mathematics, physics, engineering, and computer science.

5.6.2 Appendix · Unified Expression of Mathematical Objects

In linear algebra and broader mathematical fields, we use similar expressions to describe different mathematical objects:

- **For matrices**: We have positive semi-definite, denoted $B \succeq 0$.
- **For numbers**: We have inequalities, denoted $b \geq 0$.
- **For functions**: We have non-negativity, denoted $f(x) \geq 0, \forall x \in$ domain.
- **For operators**: We have positivity, expressed in functional analysis as $\langle Tx, x \rangle \geq 0$.

This unity demonstrates the abstract power of mathematics: different mathematical objects can share similar structures and properties.

Chapter 6

The Mathematical Language of Nature—Fractals

6.1 What Are Fractals?

Fractals are geometric shapes with self-similarity, exhibiting similar structures at different scales. Classic examples:

- Koch snowflake curve
- Mandelbrot set
- Sierpinski triangle

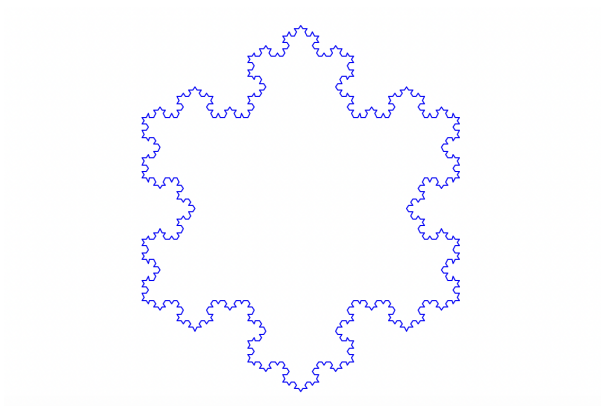


Figure 1: Koch snowflake curve

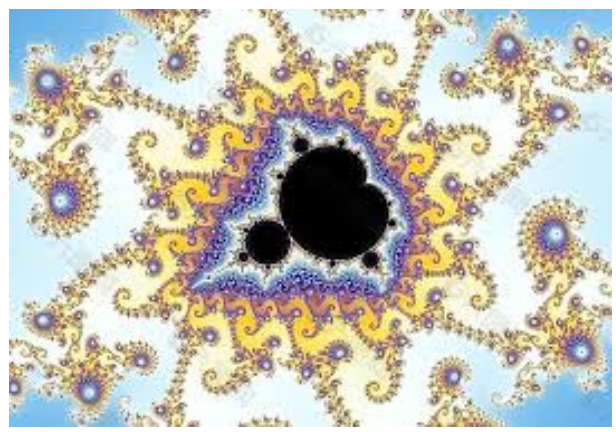


Figure 2: Mandelbrot set

6.2 Construction and Properties of the Koch Snowflake

The Koch snowflake is generated by a simple iterative process:

1. Start with an equilateral triangle.
2. Divide each side into three equal segments.
3. Construct an equilateral triangle on the middle segment, pointing outward.
4. Remove the base of the new triangle (the original middle segment).
5. Repeat steps 2-4 infinitely.

After each iteration, the perimeter increases by a factor of $\frac{4}{3}$, but the area converges to a finite value. Let the initial triangle have side length s.

Perimeter Growth

Initial perimeter: $P_0 = 3s$
After first iteration: Each side becomes 4 smaller segments of length $s/3$, so

$$P_1 = 3 \times 4 \times \frac{s}{3} = 4s = P_0 \times \frac{4}{3}$$

After n iterations:

$$P_n = P_0 \times \left(\frac{4}{3}\right)^n = 3s \times \left(\frac{4}{3}\right)^n$$

As $n \to \infty$, $P_n \to \infty$.

Area Convergence

Initial area: $A_0 = \frac{\sqrt{3}}{4}s^2$
First iteration adds 3 small triangles, each with side length $s/3$:

$$\Delta A_1 = 3 \times \frac{\sqrt{3}}{4}\left(\frac{s}{3}\right)^2 = \frac{\sqrt{3}}{4}s^2 \times \frac{1}{3}$$

Second iteration adds $3 \times 4 = 12$ even smaller triangles:

$$\Delta A_2 = 12 \times \frac{\sqrt{3}}{4}\left(\frac{s}{9}\right)^2 = \frac{\sqrt{3}}{4}s^2 \times \frac{4}{27}$$

Number of triangles added at the n-th iteration: $3 \times 4^{n-1}$
Area added:

$$\Delta A_n = 3 \times 4^{n-1} \times \frac{\sqrt{3}}{4}\left(\frac{s}{3^n}\right)^2 = \frac{\sqrt{3}}{4}s^2 \times \frac{4^{n-1}}{9^{n-1}} \times \frac{1}{3}$$

Total area is a geometric series sum:

$$A_\infty = A_0 + \sum_{n=1}^{\infty} \Delta A_n = \frac{\sqrt{3}}{4}s^2\left(1 + \frac{1}{3}\sum_{n=0}^{\infty}\left(\frac{4}{9}\right)^n\right)$$

Using the geometric series formula $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ (for $|r| < 1$):

$$A_\infty = \frac{\sqrt{3}}{4}s^2\left(1 + \frac{1}{3} \times \frac{1}{1 - \frac{4}{9}}\right) = \frac{\sqrt{3}}{4}s^2\left(1 + \frac{1}{3} \times \frac{9}{5}\right)$$

$$A_\infty = \frac{\sqrt{3}}{4}s^2\left(1 + \frac{3}{5}\right) = \frac{\sqrt{3}}{4}s^2 \times \frac{8}{5} = \frac{2\sqrt{3}}{5}s^2$$

Or expressed as:

$$A_\infty = \frac{8}{5}A_0$$

where A_0 is the area of the initial triangle.

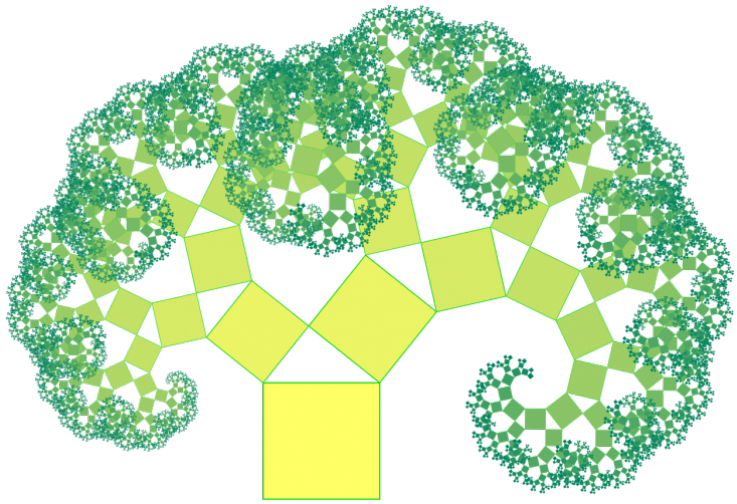


Figure 3: ”Pythagoras” tree, resembling a vegetable

6.3 Fractal Dimension

Unlike traditional integer dimensions, fractals have fractional dimensions. The dimension of the Koch curve can be found via the similarity dimension formula:

$$D = \frac{\log N}{\log s}$$

where N is the number of smaller copies after each iteration, and s is the scaling factor.

For the Koch curve: each iteration divides a side into 4 smaller segments, each $1/3$ the original length ($s = 3$). Thus:

$$D = \frac{\log 4}{\log 3} \approx 1.2618$$

This means the Koch curve ”fills space” more than a 1D line but not as much as a 2D plane.

6.4 Fractals in Nature

Fractals exist not only in mathematics, but are pervasive patterns in nature:

- **Plant growth**: Ferns, broccoli, tree branching exhibit fractal characteristics.
- **Mountain contours**: Observing mountains at different scales reveals similar irregular shapes.
- **Cloud edges**: Cloud boundaries show similar irregularity at different magnifications.
- **Vascular branching**: The branching pattern from arteries to capillaries has self-similarity.

Figure 4: Plant growth

Figure 5: Lightning pattern

6.5 The Significance of Fractals

There are many more examples of fractals; for further exploration, refer to the interesting book *The Fractal Geometry of Nature*.

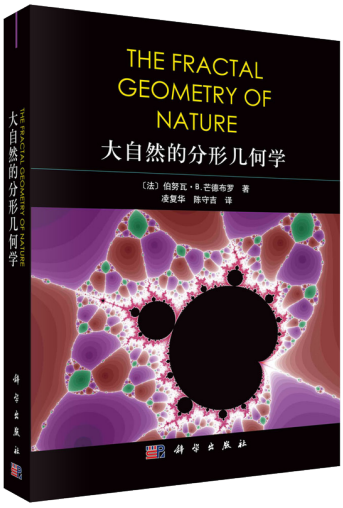


Figure 6: *The Fractal Geometry of Nature*

The significance of fractal geometry lies not only in its mathematical beauty but also in providing a new language to describe complex natural phenomena:

- **Describing Complexity**: Can describe irregular, fragmented shapes difficult for traditional Euclidean geometry.
- **Natural Optimization**: Many natural fractal structures (e.g., blood vessels) result from space-filling and efficiency optimization.
- **Art-Science Integration**: Fractal patterns have wide applications in computer graphics, art design, architecture, etc.

Fractal theory shows that simple iterative rules can produce extremely complex structures. This is not only a mathematical marvel but also a universal principle of nature's creativity and diversity.

Closing Remarks: Cultivate Your TASTE

In the age of AI, a question lingers in the minds of technical workers: ”Which professions or skills will not be replaced by AI?” I am not well-versed enough to answer this definitively. But I’d like to share a story that might provide some inspiration for pondering this question.

The renowned scientist Chen-Ning Yang often mentions a word: ”taste”. You can roughly understand it as the sum of a person’s preferences, interests, aesthetics, judgment, appreciation, and personal aspirations. **Being inexplicably moved by something, feeling an impulsive urge to explore a problem—these are driven by taste.**

It is said that once, a 15-year-old prodigy wanted to join Dr. Yang’s research institute. During the interview, Yang asked several quantum mechanics questions, which the youth answered correctly. But then Yang asked another question: ”Among these quantum mechanics problems, which one do you find beautiful?”

The youth couldn’t answer.

Dr. Yang later commented: ”Although he had absorbed a lot, he hadn’t developed a taste. If someone studies quantum mechanics and doesn’t feel that some parts are important, some are beautiful, some are worth arguing about vehemently, then I think he hasn’t truly internalized the subject.” Reportedly, Yang did not accept the student.

You see, this 15-year-old was smart and knowledgeable, but perhaps he lacked something: what Dr. Yang called ”taste.” Taste, combined with ability, temperament, and opportunity, determines a person’s style, and style determines contribution.

In other words, your judgment shapes your work.

So where does taste come from? Yang said it’s related to personal ability, family environment, early education, personality, and luck. In essence, we can understand taste as something accumulated over time. The works you’ve seen, projects you’ve done, pitfalls you’ve encountered—slowly form your taste.

Today’s AI is powerful, but it lacks what we call ”taste.” AI can draw, but it doesn’t know *what* to draw. It can write, but it doesn’t know *what* to write. It can solve problems, but it doesn’t know *which* problems are beautiful.

Humans are born with their own preferences, aesthetics, and ideas. For example, someone with 10 years of animation experience can instantly tell if the pacing of a scene is off. A designer with 10 years of experience can immediately see if a color combination is uncomfortable. In programming, with the rise of ”ambient programming,” anyone can have AI write code. But what constitutes good code? What architectures are reasonable? What code is efficient? These judgments likely require the taste of an experienced programmer.

This article of mine might represent my taste in mathematics. Two and a half years is a short time; what I’ve learned may not be profoundly sophisticated.

From natural numbers to reals to complexes, from finite arithmetic to limit theory, and from simple geometry to complex fractals, the development of mathematics begins with simple questions and reaches profound conclusions. The beauty of mathematics lies in this logical purity: a few simple axioms, through rigorous reasoning, can construct vast systems that explain the laws of the universe. The real number system provides the foundation for analysis, the complex system offers detour tactics for computation, and fractals show us how simple rules can generate incredibly complex structures. Mathematics not only describes the world but also creates new ways of understanding it.